TwinShield: Static Conditional Admission Assurance for LLM Routing under Imperfect Network Twins

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***Abstract*—Validation against a stale network digital twin can certify a plan that fails after deployment. We study a static interval gate that admits an LLM-generated unsplittable-routing plan only when capacity and latency hold throughout a declared error envelope. On 128 tasks selected before hidden physical states were generated, one chronologically fixed service-model run improves from 78 gate passes to 103 after two counterexample revisions; matched generic robust instructions reach 102 and rejection-only feedback 93. Across five additional complete runs, specific-minus-generic differences are $+1, +1, -2, -1, -1, 0$ tasks: exact violations have no consistent advantage. A fixed-envelope full-graph MILP and a separate bisection max-headroom solver both certify 103 tasks in milliseconds. Common-inventory stress preserves the same out-of-envelope failure pattern. Adding a 10% worst-case demand envelope reduces joint-certificate coverage to 31/128 for both feedback variants and 34/128 for the MILP. Inside-envelope safety is a deterministic invariant, not a field-risk estimate. On these small formalized instances, deterministic optimization should replace LLM route construction; broader LLM roles remain an untested architectural hypothesis.**

I. Introduction

LLM network agents increasingly combine planning with network digital twins (NDTs) and executable validation [1], [2], [3]. Validation against a stale twin, however, can create *false assurance*: a machine-checkable pass for a plan that violates the physical network. NDT synchronization and fidelity remain open architectural concerns [4], [5].

We isolate this mismatch in static unsplittable traffic engineering (TE). An LLM receives estimated capacities/latencies and three flow intents, proposes one path per flow, and is admitted only after a fail-closed interval check. This paper contributes: (i) a conditional admission/revision pattern with a conservative numerical audit; (ii) a hidden-state selection audit and exact plan lineage; (iii) six complete trajectories with equal-call information controls and exact finite-array contrasts; and (iv) complete physical, common-inventory, and demand-envelope sensitivities against deployable full-graph solvers. The distinctive contribution is this experimental audit, not a new shielding or optimization primitive.

Related work. NetConfEval and NetArena execute LLM-generated network artifacts/actions [6], [7]; Cornetto, Aether, a Kubernetes/NDT VNF agent, and retry-validated SDN flow generation combine agents with checking or repair [8], [9], [3], [10], [11]. TypoNet separates translation from deterministic reasoning [12], and robust routing predates LLM agents [13], [14]. Simplex assurance and shielding interpose a verified mechanism between an untrusted controller and actuation [15], [16]. TwinShield instantiates that pattern for NDT mismatch and studies admission, selection, run sensitivity, and envelope failure rather than claiming a new robust-optimization method.

II. Conditional Certificate and Calibration

Let visible twin $\hat{G} = (V, E, \hat{c}, \hat{\ell})$ assign positive capacity $\hat{c}_e$ and latency $\hat{\ell}_e$. Flow $i = (s_i, t_i, d_i, L_i)$ has fixed positive demand d_i and latency limit L_i. A plan P assigns each flow a simple, endpoint-correct path p_i. With $q_e = \sum_{i:e\in p_i} d_i$, physical state (c, ℓ) is valid iff

$$q_e \le c_e \ \forall e, \qquad \sum_{e\in p_i} \ell_e \le L_i \ \forall i. \tag{1}$$

For N intended tasks we report admitted/safe/unsafe/no-deployment counts (A, S, U, X), where $A = S + U$ and $X = N - A$. Unsafe risk is U/A for $A > 0$ and NA otherwise. The artifact additionally separates validator abstention from execution failure inside X.

For declared $\epsilon \in [0, 1)$, the gate checks Eq. (1) after substituting

$$c_e^- = (1-\epsilon)\hat{c}_e, \qquad \ell_e^+ = (1+\epsilon)\hat{\ell}_e. \tag{2}$$

Proposition 1 (conditional admission). Fix demands and a valid simple-path plan. If the gate passes under Eq. (2), comparisons are conservative, and the physical state obeys $c_e \ge c_e^-$ and $0 < \ell_e \le \ell_e^+$, then P satisfies Eq. (1). *Proof.* Each physical capacity is no smaller than the checked lower bound, and each path's physical latency is termwise no larger than its checked upper-bound sum. □

Thus zero bounded-law failures at $\epsilon = .2$ check implementation, not a probability. A nominally valid fixed plan's scalar headroom is

$$h(P) = \min\left\{ \min_{e:q_e>0}\left(1 - \frac{q_e}{\hat{c}_e}\right), \min_i\left(\frac{L_i}{\sum_{e\in p_i}\hat{\ell}_e} - 1\right)\right\}, \tag{3}$$

clipped to $[0, 1)$. The LLM is asked only to pass; the deterministic comparator can optimize h separately. If demands may simultaneously rise by at most ρ, the joint gate replaces the capacity test by $(1+\rho)q_e \le (1-\epsilon)\hat{c}_e$; the latency test is unchanged.

Prospective calibration. For used-link set $E(P) = \cup_i p_i$, define the plan-specific residual

$$R_t(P) = \max_{e \in E(P)} \left\{ \left[\frac{\hat{c}_{e,t} - c_{e,t}}{\hat{c}_{e,t}} \right]_+, \left[\frac{\ell_{e,t} - \hat{\ell}_{e,t}}{\hat{\ell}_{e,t}} \right]_+ \right\}. \quad (4)$$

This avoids letting unused links dominate. Freeze the selector and one plan/twin/traffic stratum, then collect held-out telemetry. Only under *iid* Bernoulli indicators $I_t = \mathbf{1}[R_t(P) > \epsilon]$ with one fixed probability p does zero exceedances give the exact one-sided bound $p \le 1 - \alpha^{1/n}$. Generic exchangeability is insufficient: if all $I_t = B \sim \text{Bernoulli}(p)$, the zero probability is $1 - p$, not $(1 - p)^n$. At 95% confidence, a 1% target needs $n = 299$; Bonferroni control needs 437 for four strata or 505 for eight. For dependence, predeclared non-overlapping block maxima may be used only when the resulting block indicators are themselves justified iid; the bound then concerns block exceedance. Otherwise no binomial claim is made. Policy/exposure/drift changes reset calibration, and insufficient evidence fails closed. This is prospective guidance, not field calibration by our synthetic data.

III. Experimental Design

Cohort and states. Eight Internet Topology Zoo structures (11–20 nodes, 14–26 links) receive synthetic capacities, latencies, and three flows [17]. Visible tasks are nominal-feasible with target maximum utilization .695–.905. The earlier retained cohort selected the first attempt whose *candidate set*—ten visible-twin latency-ordered simple paths per flow—contained a feasible assignment in *all eight* hidden states (two error modes times four levels). Our primary *visible-only* cohort instead fixes attempt zero for the same 8×16 identifiers before generating physical states and never replaces a task. Table I exposes this selection effect. The topology list and 16 identifiers were frozen before the original main outcomes; the reviewer- requested attempt-zero rule was added later but frozen before its own physical states and model outcomes. Each base has diffuse/hotspot capacity decreases and latency increases at $\delta \in \{0, .1, .2, .3\}$.

Generator. Each logical link draws per-link capacity from $\{80, 100, 120, 140, 160\}$ (multiplied by collapsed parallel-link count). Latency is geodesic distance/200 plus $U[.15, .75]$ ms, lower-bounded by .5 ms; links without coordinates use $U[2, 8]$ ms. Endpoint pairs have hop distance ≥ 2, are seed-shuffled, and prefer disjoint endpoints. Provisional demands are $U[.8, 1.2]$; each SLO is the shortest candidate latency times $U[1.2, 1.5]$. A nominal oracle then scales all demands to seeded target utilization $U[.70, .80]$; the audited .695–.905 band absorbs rounding and solver tolerance. Seeds, source hashes, rejection reasons, and all physical states are archived.

Model policies. The chronological visible-only high-reasoning `gpt-5.6-luna` run is fixed as primary trajectory 0. Five new complete trajectories use the same 128 inputs. *Direct* deploys any structured initial plan; *Nominal gate* admits it at $\epsilon = 0$; *Robust gate* uses $\epsilon = .2$. After a parsable robust rejection, three matched policies each issue exactly two sequential stateless calls: exact counterexample violations (specific), generic robust constraints, or only the fact of rejection. Call 2 receives the last rejected candidate; after call-1 success it restarts from the original rejected context. Both calls are always stored, and the earliest exact-gate success is selected afterward. Calls use structured output, `store=false`, no conversation/previous-response ID, no API seed, and omitted temperature/top-p; this does not establish provider-side statistical independence. All incomplete/schema-invalid calls remain in intention-to-treat (ITT) denominators.

TABLE I
Selection audit. "All-8" means candidate-set feasible in both modes at all four mismatch levels; it is an outcome, not a primary selection rule.

Cohort	N	Robust MILP	All-8
All attempted	615	482 (78.4%)	128 (20.8%)
Retained	128	121 (94.5%)	128 (100%)
Visible-only	128	103 (80.5%)	28 (21.9%)

The shared system instruction requires exactly one simple, endpoint-correct route/flow using only listed edges and satisfying aggregate capacity and end-to-end latency. The user-task templates are: specific, "repair every validator violation" plus exact violations and validator; generic, "make all aggregate capacity and end-to-end latency constraints pass" plus validator but no violations; rejection-only, "rejected; produce a different feasible plan" without validator details. All include twin, flows, prior plan, identical schema/model/reasoning, and two calls. Payload length and semantics are not matched, so this compares complete information bundles, not wording or token count alone; exact templates are in the artifact.

The estimand is the fixed $6 \times 8 \times 16$ trajectory/topology/identifier array. We report every paired task difference and topology aggregate without a confidence interval: neither the six mutable-service trajectories nor the eight named graphs are asserted to be independent samples from a superpopulation.

Complete full-graph MILP. For directed arcs uv, let $x_{iuv} \in \{0, 1\}$ and $0 \le z \le 1$. Phase 1 minimizes z subject to

$$\begin{aligned} \sum_v (x_{iuv} - x_{ivu}) &= \mathbf{1}[u = s_i] - \mathbf{1}[u = t_i], \\ \sum_v x_{iuv} \le 1, \quad \sum_v x_{ivu} &\le 1, \\ \sum_v x_{ivs_i} = 0, \quad \sum_v x_{it_iv} &= 0, \\ \sum_i d_i (x_{iuv} + x_{ivu}) &\le z(1 - \epsilon)\hat{c}_{\{u,v\}}, \\ \sum_{uv} (1 + \epsilon)\hat{\ell}_{uv} x_{iuv} &\le L_i. \end{aligned} \quad (5)$$

The degree inequalities apply at every node. Phase 2 adds $z \le z^* + 10^{-8}$ and minimizes $\sum_{i,uv} (\hat{\ell}_{uv}/L_{\max} + 10^{-9} \operatorname{rank}(uv)) x_{iuv}$; positive costs eliminate disconnected cycles and the rank breaks ties. SciPy 1.18.1/HiGHS 1.12.0 uses presolve, zero MIP gap, its recorded default 10^{-6} feasibility tolerance, and 10 s/phase. Every decoded plan is then audited

with 50-digit Decimal arithmetic and must retain at least 10^{-9} relative capacity and latency slack; otherwise it fails closed. Because $z \leq 1$ is explicit, a status-2 return is solver-infeasible, never an accepted overloaded optimum.

This fixed-ϵ two-phase program is capacity-first and is *not* identified with Eq. (3). A separate solver computes

$$h^* = \sup\{\eta \in [0, .999] \mid \mathcal{F}(\eta) \neq \varnothing\}, \tag{6}$$

where $\mathcal{F}(\eta)$ is Eq. (5)'s feasible set with $\epsilon = \eta$. It performs 12 feasibility bisections, retains the plan at the feasible lower endpoint, and therefore brackets h^* within $.999/2^{12} = .000244$ unless right-censored. Recomputed plan headroom must certify that endpoint.

Misspecification and demand. Fixed primary plans receive 128 common-random draws/base/law. Bounded residuals are independent $U[0, .2]$. For correlation, one $G = .5\,\mathrm{Beta}(2, 5)$ is shared across every link and both metrics in a draw, while each metric/link has independent $U_{me} \sim U[0, .2]$ and residual $\min(.75, .8G + .2U_{me})$. Heavy-tail terms are independent $\min(.75, .07|T_3|)$ by metric/link. The revised comparison restricts all methods to the same base IDs with all five stored plan types, reports inventory loss separately, and gives raw frequencies over the fixed draw array—not confidence intervals or field risk. A post-review sensitivity re-gates fixed plans for joint $\epsilon = .2$ and $\rho \in \{0, .025, .05, .075, .1, .15, .2\}$, then replays both modes at $\delta \leq .2$. It does not rerun the LLM.

Compute. The Luna amendment made 2,472 calls, 8.709M total tokens, and cost an estimated USD 5.9248 at prices checked 2026-08-31. The local immutable `Qwen2.5-3B-Instruct-4bit` revision `4f83f8f146fdf28b512a06562b671d7af4fab457` is only a lower-bound failure illustration, not a competitive baseline. On an Apple M1 Pro/16 GB it loads in .903 s, peaks at 2.786 GB, and spends 3,642.207 s generating 714 calls.

IV. Results

Table II unifies primary and run-sensitivity results. Specific revision's 103 primary deployments are exactly the 103 MILP-feasible task IDs. Specific-minus-generic differences are $+1, +1, -2, -1, -1, 0$ tasks ($-2/768 = -.26$ points overall): two trajectory-task units are specific-only and four generic-only. By topology, only Claranet and Internet2 have net -1; the other six are zero. Specific is a subset of the fixed 103 MILP-feasible IDs in every run and equals that set in trajectories 0, 1, and 5; trajectories 2–4 miss 3, 2, and 1 IDs. Specific-minus-rejection-only is respectively $+10, +11, +15, +13, +8, +9$ tasks. Thus robust validator information helps in this array, but exact violations have no consistent advantage over generic constraints.

Table III follows the same primary plans into every physical state. At .2, all robust admissions are safe by Proposition 1; Direct and Nominal expose 38/35 diffuse and 19/16 hotspot failures. At hotspot .3, unsafe risk is 71.8% for gate, 71.8% specific/MILP, 71.6% generic, and 72.0% rejection-only: the certificate cannot repair a false envelope. The specific/generic/rejection-only X totals contain respectively 6/4/1 execution failures; all other nondeployments there are validator abstentions. No unit is discarded.

Zero-exceedance arithmetic (iid Bernoulli only)

Simultaneous K	α/K	upper at $n = 128$	n for $\leq 1\%$
1	.05000	2.31%	299
4	.01250	3.37%	437
8	.00625	3.89%	505

Latency, solver status, and model sensitivity. Primary gate median/p95 latency is 7.14/16.30 s. Specific, generic, and rejection-only pipelines are 9.39/127.09, 9.39/132.43, and 9.39/56.09 s, costing USD .472/.548/.375 as counterfactual complete policies. Visible-only robust MILP is 3.89/10.83 ms and bisection max-headroom 44.71/86.69 ms. All 128 searches pass the audit: no right censoring, maximum bracket width .000244, minimum recomputed plan-headroom excess over its lower endpoint 2.0×10^{-6}, and 103 plans certify .2. Across all 615 fixed-envelope attempts, 482 are optimal two-phase solutions and 133 are HiGHS status-2 infeasible; no timeout, decode, or Decimal-audit failure occurs. Qwen yields 0/128 gate/revision passes in each cohort, a small quantized failure case rather than a parity comparison.

Common inventory and demand. Available fixed-plan counts are 125/103/102/93/128 for initial/specific/generic/rejection-only/max-headroom; their intersection has 90 IDs. At threshold .2, heavy-tail raw unsafe frequency on this common set is 10.36% for the 78 admitted initial candidates, and 10.67/10.76/10.76/10.57% for 90 admitted plans of the other methods. Thus the earlier pattern is not only an inventory artifact, but these synthetic draws do not estimate field risk.

With joint demand inflation $\rho = .05$, admission is 52/66/65/62/68 for gate/specific/generic/rejection-only/MILP; at $\rho = .10$ it is 25/31/31/28/34, and at .20 it is zero. Every jointly admitted plan is safe over both modes and all six within-envelope state/base replays. In contrast, holding the original $\rho = 0$ admissions fixed at $\rho = .10$ makes 10.9–12.1% of those replays unsafe. Demand uncertainty therefore belongs in the declared certificate; this analysis measures fixed-plan headroom, not LLM adaptation.

V. Limitations and Conclusion

The study is synthetic and static, with eight topology structures, three flows, one-sided residuals, and no queueing, failures, adversarial telemetry, device translation, or live NDT synchronization. Visible-only selection avoids hidden-state filtering but remains generator/difficulty-conditioned. Luna is mutable; six runs are not a model population. Qwen is immutable but smaller and quantized. Stress laws and uniform demand inflation are not traces; calibration is prospective. MILP timing on 11–20-node, 14–26-link, three-flow instances is not production-scale evidence.

Within those limits, TwinShield provides static conditional assurance: safety holds only inside a declared and conservatively checked interval. The empirical questions are admission, selection, run variability, and failure outside the premise.

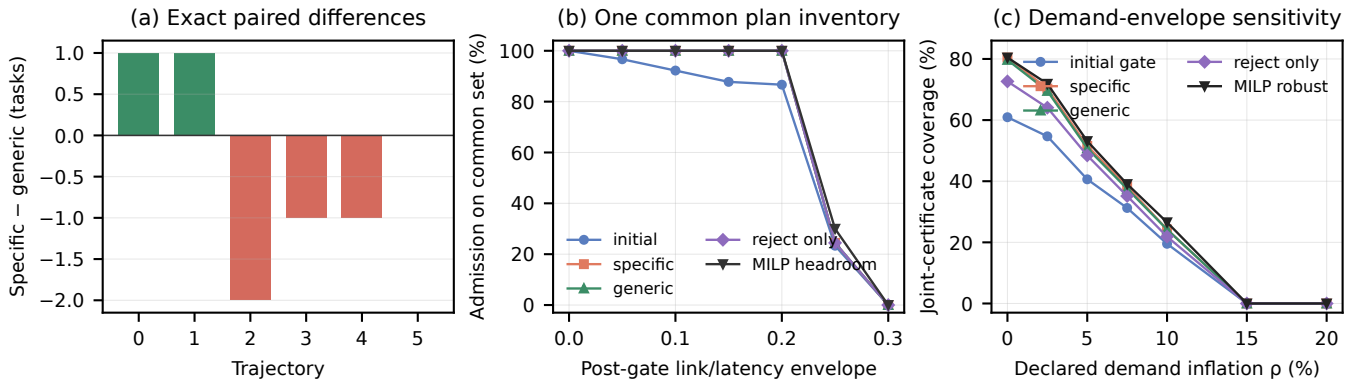


Fig. 1. Post-review soundness analyses. (a) All six paired specific-minus-generic differences. (b) Threshold admission on one common 90-base inventory; the law label is immaterial to admission. (c) Fixed-plan joint link/latency/demand envelope. No curve is a population confidence interval, and no missing plan is reconstructed.

TABLE II
ROBUST-VALID /128. RUN COLUMN IS DESCRIPTIVE MEDIAN [IQR]; RANGE OVER ALL SIX TRAJECTORIES. REMAINING COLUMNS ARE PRIMARY COUNTS /16 BY TOPOLOGY.

Policy	Primary	Six-run summary	Abi	Cla	Ibm	Int2	Nav	Nsf	P1	Spr
Robust gate	78	73.5 [70.5,77.3]; 70–78	6	10	8	9	8	14	13	10
Specific	103	102.5 [101.3,103]; 100–103	12	14	9	13	14	14	13	14
Generic	102	102 [102,102.8]; 102–103	12	14	9	13	13	14	13	14
Reject only	93	92.5 [89,93.8]; 85–94	10	14	9	11	11	14	13	11
Robust MILP	103	—	12	14	9	13	14	14	13	14

TABLE III
UNIFIED VISIBLE-ONLY PRIMARY OUTCOMES $(A/S/U/X)$, $N = 128$; D/H DENOTES DIFFUSE/HOTSPOT. X IS NO DEPLOYMENT (ABSTENTION OR EXECUTION FAILURE).

Method	D,.2	H,.2	D,.3	H,.3
Direct	128/90/38/0	128/109/19/0	128/52/76/0	128/32/96/0
Nominal gate	125/90/35/3	125/109/16/3	125/52/73/3	125/32/93/3
Robust gate	78/78/0/50	78/78/0/50	78/50/28/50	78/22/56/50
Specific	103/103/0/25	103/103/0/25	103/58/45/25	103/29/74/25
Generic	102/102/0/26	102/102/0/26	102/58/44/26	102/29/73/26
Reject only	93/93/0/35	93/93/0/35	93/55/38/35	93/26/67/35
Robust MILP	103/103/0/25	103/103/0/25	103/59/44/25	103/29/74/25

Generic robust feedback matches exact counterexamples, while rejection-only feedback loses coverage. Once intent is formal, the full-graph MILP reaches the same primary ceiling in milliseconds and should replace LLM route construction on these evaluated instances. Assigning LLMs to intent translation or explanation is a related-work-supported architectural hypothesis, not an experimental finding here. Deployment additionally needs held-out, selection-aware calibration, a demand envelope, and drift resets.

APPENDIX A
AI RESEARCH DISCLOSURE

The operative initial human request was: create the entire track paper end-to-end—select/scope the problem, run experiments, plot, make figures, and write—using this Mac, optional agents, optional Hetzner (about USD 30), and the stated OpenAI credit. Codex desktop `gpt-5.6-sol` selected the question and produced every technical edit. It used local files/shell, Python/LaTeX, browser/web research, deterministic validators/solvers, and the OpenAI API; Hetzner was unused. Humans supplied only objectives, access, review links, page/budget constraints, and continuation/publication approvals, never technical text or results. The exact prompt is provenance/initial_prompt.md, SHA-256 `3b23d5b1643e8b4a9d09b5bb8dbdeefe3da2a3a07710bf77b5b9ceadcef68200`; the full ledger is provenance/human_action_ledger_v6.md, SHA-256 `6ca654d899e4190a2d3fb3b261e08cd8b28d782f33dec279987bbd3db7bdaff1`.

Plan generation used stateless `gpt-5.6-luna` structured requests and local greedy Qwen; three inherited-model subagents assisted only the initial audit. For each revision Codex preserved the prior version, archived supplied AI critiques, mapped concerns, froze any new experiment before outcomes, kept failed calls/IDs, ran objective tests, and regenerated the paper. Eight public AI-review conversations were critique inputs, not peer review. The canonical five-round index is reviews/review_round_registry_v6.md, SHA-256 `cc2b340a7f05b7a823ca03a4b4e6ac50bfc3f828a419fea145c2349466639788`; it records exact reviewed PDF hashes, visible prompts/responses, message IDs, and page model slugs. Seven used the venue-specific request and one assumed an archival track. Hidden reasoning is unavailable.

Cumulative estimated OpenAI API spend is USD 19.6825; v6 added zero calls/cost, and Qwen/MILP/Hetzner API cost is zero. Credentials, account IDs, hidden reasoning, quarantined transport failures, and Qwen weights are excluded. The v5 raw-service release remains at https://github.com/trondvatt/twinshield-cnsm2026/releases/tag/v5. The companion submission file is `twinshield-cnsm2026-artifact-v6.zip`, SHA-256 `5ee4d7a75fba5fa28fc794076fa4c33ff138f21590fb2c635e5ba9163d868c48`; the PDF is separate to avoid circular hashing.

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